SIM1003 Calculus II
Tutorial 3

All even-numbered questions in the following exercises

EXERCISES 8.5

Expanding Quotients into Partial Fractions

Expand the quotients in Exercises 1–8 by partial fractions.

1. $\dfrac{5x - 13}{(x - 3)(x - 2)}$

2. $\dfrac{5x - 7}{x^2 - 3x + 2}$

3. $\dfrac{x + 4}{(x + 1)^2}$

4. $\dfrac{2x + 2}{x^2 - 2x + 1}$

5. $\dfrac{z + 1}{z^2(z - 1)}$

6. $\dfrac{z}{z^3 - z^2 - 6z}$

7. $\dfrac{t^2 + 8}{t^2 - 5t + 6}$

8. $\dfrac{t^4 + 9}{t^4 + 9t^2}$

Nonrepeated Linear Factors

In Exercises 9–16, express the integrand as a sum of partial fractions and evaluate the integrals.

9. $\displaystyle\int \frac{dx}{1 - x^2}$

10. $\displaystyle\int \frac{dx}{x^2 + 2x}$

11. $\displaystyle\int \frac{x + 4}{x^2 + 5x - 6}\,dx$

12. $\displaystyle\int \frac{2x + 1}{x^2 - 7x + 12}\,dx$

13. $\displaystyle\int_4^8 \frac{y\,dy}{y^2 - 2y - 3}$

14. $\displaystyle\int_{1/2}^1 \frac{y + 4}{y^2 + y}\,dy$

15. $\displaystyle\int \frac{dt}{t^3 + t^2 - 2t}$

16. $\displaystyle\int \frac{x + 3}{2x^3 - 8x}\,dx$

Repeated Linear Factors

In Exercises 17–20, express the integrand as a sum of partial fractions and evaluate the integrals.

17. $\displaystyle\int_0^1 \frac{x^3\,dx}{x^2 + 2x + 1}$

18. $\displaystyle\int_{-1}^0 \frac{x^3\,dx}{x^2 - 2x + 1}$

19. $\displaystyle\int \frac{dx}{(x^2 - 1)^2}$

20. $\displaystyle\int \frac{x^2\,dx}{(x - 1)(x^2 + 2x + 1)}$

Irreducible Quadratic Factors

In Exercises 21–32, express the integrand as a sum of partial fractions and evaluate the integrals.

21. $\displaystyle\int_0^1 \frac{dx}{(x + 1)(x^2 + 1)}$

22. $\displaystyle\int_1^{\sqrt{3}} \frac{3t^2 + t + 4}{t^3 + t}\,dt$

23. $\displaystyle\int \frac{y^2 + 2y + 1}{(y^2 + 1)^2}\,dy$

24. $\displaystyle\int \frac{8x^2 + 8x + 2}{(4x^2 + 1)^2}\,dx$

25. $\displaystyle\int \frac{2s + 2}{(s^2 + 1)(s - 1)^3}\,ds$

26. $\displaystyle\int \frac{s^4 + 81}{s(s^2 + 9)^2}\,ds$

27. $\displaystyle\int \frac{x^2 - x + 2}{x^3 - 1}\,dx$

28. $\displaystyle\int \frac{1}{x^4 + x}\,dx$

29. $\displaystyle\int \frac{x^2}{x^4 - 1}\,dx$

30. $\displaystyle\int \frac{x^2 + x}{x^4 - 3x^2 - 4}\,dx$

31. $\displaystyle\int \frac{2\theta^3 + 5\theta^2 + 8\theta + 4}{(\theta^2 + 2\theta + 2)^2}\,d\theta$

32. $\displaystyle\int \frac{\theta^4 - 4\theta^3 + 2\theta^2 - 3\theta + 1}{(\theta^2 + 1)^3}\,d\theta$

Improper Fractions

In Exercises 33–38, perform long division on the integrand, write the proper fraction as a sum of partial fractions, and then evaluate the integral.

33. $\displaystyle\int \frac{2x^3 - 2x^2 + 1}{x^2 - x}\,dx$

34. $\displaystyle\int \frac{x^4}{x^2 - 1}\,dx$

35. $\displaystyle\int \frac{9x^3 - 3x + 1}{x^3 - x^2}\,dx$

36. $\displaystyle\int \frac{16x^3}{4x^2 - 4x + 1}\,dx$

37. $\displaystyle\int \frac{y^4 + y^2 - 1}{y^3 + y}\,dy$

38. $\displaystyle\int \frac{2y^4}{y^3 - y^2 + y - 1}\,dy$

Evaluating Integrals

Evaluate the integrals in Exercises 39–54.

39. $\displaystyle\int \frac{e^t\,dt}{e^{2t} + 3e^t + 2}$

40. $\displaystyle\int \frac{e^{4t} + 2e^{2t} - e^t}{e^{2t} + 1}\,dt$

41. $\displaystyle\int \frac{\cos y\,dy}{\sin^2 y + \sin y - 6}$

42. $\displaystyle\int \frac{\sin\theta\,d\theta}{\cos^2\theta + \cos\theta - 2}$

43. $\displaystyle\int \frac{(x-2)^2\tan^{-1}(2x) - 12x^3 - 3x}{(4x^2+1)(x-2)^2}\,dx$

44. $\displaystyle\int \frac{(x+1)^2\tan^{-1}(3x) + 9x^3 + x}{(9x^2+1)(x+1)^2}\,dx$

45. $\displaystyle\int \frac{1}{x^{3/2} - \sqrt{x}}\,dx$

46. $\displaystyle\int \frac{1}{(x^{1/3} - 1)\sqrt{x}}\,dx$

(*Hint:* Let $x = u^6$.)

47. $\displaystyle\int \frac{\sqrt{x+1}}{x}\,dx$

(*Hint:* Let $x + 1 = u^2$.)

48. $\displaystyle\int \frac{1}{x\sqrt{x+9}}\,dx$

49. $\displaystyle\int \frac{1}{x(x^4+1)}\,dx$

$\left(\textit{Hint: } \text{Multiply by } \dfrac{x^3}{x^3}.\right)$

50. $\displaystyle\int \frac{1}{x^6(x^5+4)}\,dx$

51. $\displaystyle\int \frac{1}{\cos 2\theta \sin\theta}\,d\theta$

52. $\displaystyle\int \frac{1}{\cos\theta + \sin 2\theta}\,d\theta$

53. $\displaystyle\int \frac{\sqrt{1+\sqrt{x}}}{x}\,dx$

54. $\displaystyle\int \frac{\sqrt{x}}{\sqrt{2-\sqrt{x}} + \sqrt{x}}\,dx$

EXERCISES 8.8

Evaluating Improper Integrals

The integrals in Exercises 1–34 converge. Evaluate the integrals without using tables.

1. $\displaystyle\int_0^{\infty} \frac{dx}{x^2 + 1}$

2. $\displaystyle\int_1^{\infty} \frac{dx}{x^{1.001}}$

3. $\displaystyle\int_0^{1} \frac{dx}{\sqrt{x}}$

4. $\displaystyle\int_0^{4} \frac{dx}{\sqrt{4 - x}}$

5. $\displaystyle\int_{-1}^{1} \frac{dx}{x^{2/3}}$

6. $\displaystyle\int_{-8}^{1} \frac{dx}{x^{1/3}}$

7. $\displaystyle\int_0^{1} \frac{dx}{\sqrt{1 - x^2}}$

8. $\displaystyle\int_0^{1} \frac{dr}{r^{0.999}}$

9. $\displaystyle\int_{-\infty}^{-2} \frac{2\,dx}{x^2 - 1}$

10. $\displaystyle\int_{-\infty}^{2} \frac{2\,dx}{x^2 + 4}$

11. $\displaystyle\int_2^{\infty} \frac{2}{v^2 - v}\,dv$

12. $\displaystyle\int_2^{\infty} \frac{2\,dt}{t^2 - 1}$

13. $\displaystyle\int_{-\infty}^{\infty} \frac{2x\,dx}{(x^2 + 1)^2}$

14. $\displaystyle\int_{-\infty}^{\infty} \frac{x\,dx}{(x^2 + 4)^{3/2}}$

15. $\displaystyle\int_0^{1} \frac{\theta + 1}{\sqrt{\theta^2 + 2\theta}}\,d\theta$

16. $\displaystyle\int_0^{2} \frac{s + 1}{\sqrt{4 - s^2}}\,ds$

17. $\displaystyle\int_0^{\infty} \frac{dx}{(1 + x)\sqrt{x}}$

18. $\displaystyle\int_1^{\infty} \frac{1}{x\sqrt{x^2 - 1}}\,dx$

19. $\displaystyle\int_0^{\infty} \frac{dv}{(1 + v^2)(1 + \tan^{-1} v)}$

20. $\displaystyle\int_0^{\infty} \frac{16 \tan^{-1} x}{1 + x^2}\, dx$

21. $\displaystyle\int_{-\infty}^{0} \theta e^{\theta}\, d\theta$

22. $\displaystyle\int_0^{\infty} 2e^{-\theta} \sin \theta\, d\theta$

23. $\displaystyle\int_{-\infty}^{0} e^{-|x|}\, dx$

24. $\displaystyle\int_{-\infty}^{\infty} 2xe^{-x^2}\, dx$

25. $\displaystyle\int_0^{1} x \ln x\, dx$

26. $\displaystyle\int_0^{1} (-\ln x)\, dx$

27. $\displaystyle\int_0^{2} \frac{ds}{\sqrt{4 - s^2}}$

28. $\displaystyle\int_0^{1} \frac{4r\, dr}{\sqrt{1 - r^4}}$

29. $\displaystyle\int_1^{2} \frac{ds}{s\sqrt{s^2 - 1}}$

30. $\displaystyle\int_2^{4} \frac{dt}{t\sqrt{t^2 - 4}}$

31. $\displaystyle\int_{-1}^{4} \frac{dx}{\sqrt{|x|}}$

32. $\displaystyle\int_0^{2} \frac{dx}{\sqrt{|x - 1|}}$

33. $\displaystyle\int_{-1}^{\infty} \frac{d\theta}{\theta^2 + 5\theta + 6}$

34. $\displaystyle\int_0^{\infty} \frac{dx}{(x + 1)(x^2 + 1)}$

Testing for Convergence

In Exercises 35–68, use integration, the Direct Comparison Test, or the Limit Comparison Test to test the integrals for convergence. If more than one method applies, use whatever method you prefer.

35. $\displaystyle\int_{1/2}^{2} \frac{dx}{x \ln x}$

36. $\displaystyle\int_{-1}^{1} \frac{d\theta}{\theta^2 - 2\theta}$

37. $\displaystyle\int_{1/2}^{\infty} \frac{dx}{x (\ln x)^3}$

38. $\displaystyle\int_{0}^{\infty} \frac{d\theta}{\theta^2 - 1}$

39. $\displaystyle\int_{0}^{\pi/2} \tan \theta \, d\theta$

40. $\displaystyle\int_{0}^{\pi/2} \cot \theta \, d\theta$

41. $\displaystyle\int_{0}^{1} \frac{\ln x}{x^2} \, dx$

42. $\displaystyle\int_{1}^{2} \frac{dx}{x \ln x}$

43. $\displaystyle\int_{0}^{\ln 2} x^{-2} e^{-1/x} \, dx$

44. $\displaystyle\int_{0}^{1} \frac{e^{-\sqrt{x}}}{\sqrt{x}} \, dx$

45. $\displaystyle\int_{0}^{\pi} \frac{dt}{\sqrt{t} + \sin t}$

46. $\displaystyle\int_{0}^{1} \frac{dt}{t - \sin t}$ (*Hint:* $t \geq \sin t$ for $t \geq 0$)

47. $\displaystyle\int_{0}^{2} \frac{dx}{1 - x^2}$

48. $\displaystyle\int_{0}^{2} \frac{dx}{1 - x}$

49. $\displaystyle\int_{-1}^{1} \ln |x| \, dx$

50. $\displaystyle\int_{-1}^{1} -x \ln |x| \, dx$

51. $\displaystyle\int_{1}^{\infty} \frac{dx}{x^3 + 1}$

52. $\displaystyle\int_{4}^{\infty} \frac{dx}{\sqrt{x} - 1}$

53. $\displaystyle\int_2^{\infty} \frac{dv}{\sqrt{v-1}}$

54. $\displaystyle\int_0^{\infty} \frac{d\theta}{1+e^{\theta}}$

55. $\displaystyle\int_0^{\infty} \frac{dx}{\sqrt{x^6+1}}$

56. $\displaystyle\int_2^{\infty} \frac{dx}{\sqrt{x^2-1}}$

57. $\displaystyle\int_1^{\infty} \frac{\sqrt{x+1}}{x^2}\,dx$

58. $\displaystyle\int_2^{\infty} \frac{x\,dx}{\sqrt{x^4-1}}$

59. $\displaystyle\int_{\pi}^{\infty} \frac{2+\cos x}{x}\,dx$

60. $\displaystyle\int_{\pi}^{\infty} \frac{1+\sin x}{x^2}\,dx$

61. $\displaystyle\int_4^{\infty} \frac{2\,dt}{t^{3/2}-1}$

62. $\displaystyle\int_2^{\infty} \frac{1}{\ln x}\,dx$

63. $\displaystyle\int_1^{\infty} \frac{e^x}{x}\,dx$

64. $\displaystyle\int_{e^e}^{\infty} \ln(\ln x)\,dx$

65. $\displaystyle\int_1^{\infty} \frac{1}{\sqrt{e^x-x}}\,dx$

66. $\displaystyle\int_1^{\infty} \frac{1}{e^x-2^x}\,dx$

67. $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{\sqrt{x^4+1}}$

68. $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{e^x+e^{-x}}$